

## MCCB – MOULDED CASE CIRCUIT BREAKER

### COMPLETE TECHNICAL NOTES, CHANGEOVER, AUTO CHANGEOVER & INTERLOCKING

#### 1. INTRODUCTION

MCCB stands for Moulded Case Circuit Breaker.

An MCCB is a low-voltage circuit breaker used for switching, protection and isolation of electrical circuits. It protects electrical systems against abnormal conditions such as overload and short circuit.

MCCBs are commonly used in:

- Main LT Panels
- PCC Panels
- MCC Panels
- Distribution Boards
- Transformer Feeders
- Generator Feeders
- Motor Feeders
- Capacitor Panels
- HVAC Panels
- Industrial Distribution Systems
- Changeover Systems

#### 2. MAIN FUNCTIONS OF MCCB

The main functions of an MCCB are:

1. Switching the electrical circuit ON and OFF.
2. Protection against overload.
3. Protection against short circuit.
4. Electrical isolation.
5. Protection against other faults when suitable protection functions are available.
6. Remote or automatic operation when suitable accessories are installed.
7. Changeover operation when used with motorized accessories and control circuits.

### 3. MCCB CONSTRUCTION

A typical MCCB consists of the following components:

- Moulded Insulated Case
- Fixed Contacts
- Moving Contacts
- Arc Extinguishing System
- Operating Handle
- Trip Mechanism
- Thermal Element
- Magnetic Element
- Electronic Trip Unit, where applicable
- Line Terminals
- Load Terminals
- Auxiliary Contacts

- Alarm Contacts
- Shunt Trip
- Under Voltage Release
- Motor Operator
- Rotary Handle, where applicable

#### 4. MCCB WORKING PRINCIPLE

During normal operation, the MCCB contacts remain closed and current flows through the circuit.

When an abnormal electrical condition occurs, the trip mechanism operates and opens the contacts.

During an overload condition, the thermal or electronic protection function detects excessive current and initiates the trip operation.

During a short-circuit condition, the magnetic or electronic protection function operates and opens the MCCB.

The exact operating characteristics depend on the selected MCCB and trip unit.

#### 5. TYPES OF MCCB

MCCBs can be classified according to their trip mechanism and application.

##### 5.1 THERMAL-MAGNETIC MCCB

A thermal-magnetic MCCB generally provides:

- Thermal overload protection
- Magnetic short-circuit protection

This type is commonly used in general low-voltage distribution systems.

## 5.2 ELECTRONIC MCCB

An electronic MCCB uses an electronic trip unit for protection.

Depending on the model, it may provide:

- Long-Time Protection
- Short-Time Protection
- Instantaneous Protection
- Ground/Earth Fault Protection

The exact functions depend on the selected manufacturer and model.

## 5.3 FIXED TRIP MCCB

A fixed-trip MCCB has predefined protection characteristics with limited or no user adjustment.

## 5.4 ADJUSTABLE TRIP MCCB

An adjustable-trip MCCB allows selected protection settings to be adjusted according to the application and protection requirements.

## 6. MCCB POLE CONFIGURATION

MCCBs are available in different pole configurations.

### 6.1 2-POLE MCCB

A 2-pole MCCB can be used for suitable single-phase applications where switching/protection of the required conductors is necessary.

### 6.2 3-POLE MCCB

A 3-pole MCCB is commonly used for three-phase systems.

### 6.3 4-POLE MCCB

A 4-pole MCCB can be used when neutral switching is required along with the three phases.

The required pole configuration depends on the system design.

## 7. MCCB CURRENT RATINGS

MCCBs are available in various current ratings.

Typical ratings may include:

16 A

25 A

32 A

40 A

63 A

80 A

100 A

125 A

160 A

200 A

250 A

320 A

400 A

500 A

630 A

800 A

1000 A

1250 A

Higher ratings may also be available depending on the manufacturer and product range.

The final MCCB rating shall be selected according to the load current, cable capacity, system conditions and protection requirements.

## 8. MCCB VOLTAGE RATING

MCCB selection shall consider the system operating voltage.

Important parameters include:

- Rated Operational Voltage –  $U_e$
- Rated Insulation Voltage –  $U_i$
- Rated Impulse Withstand Voltage –  $U_{imp}$

The selected MCCB shall have suitable voltage ratings for the electrical system.

## 9. MCCB BREAKING CAPACITY

Breaking capacity is one of the most important parameters of an MCCB.

### 9.1 $I_{cu}$

$I_{cu}$  means Ultimate Short-Circuit Breaking Capacity.

It represents the maximum prospective short-circuit current that the MCCB is capable of interrupting under specified test conditions.

## 9.2 Ics

Ics means Service Short-Circuit Breaking Capacity.

It represents the service breaking performance of the MCCB according to the applicable standard and manufacturer's specification.

The required breaking capacity shall be selected according to the prospective short-circuit current at the installation point.

## 10. MCCB PROTECTION FUNCTIONS

Depending on the MCCB type, typical protection functions may include:

- Overload Protection
- Short-Circuit Protection
- Instantaneous Protection
- Short-Time Protection
- Earth/Ground Fault Protection
- Under Voltage Protection through accessories
- Other protection functions depending on the electronic trip unit

Exact functions depend on the selected MCCB model.

## 11. MCCB TRIP UNIT

The trip unit detects abnormal electrical conditions and initiates the opening of the MCCB.



Common trip unit types are:

- Thermal-Magnetic Trip Unit
- Electronic Trip Unit

Electronic trip units may provide adjustable protection functions.

The exact adjustment ranges depend on the MCCB manufacturer and model.

## 12. MCCB ACCESSORIES

Common MCCB accessories include:

1. Auxiliary Contact
2. Alarm Contact
3. Shunt Trip
4. Under Voltage Release
5. Motor Operator
6. Rotary Handle
7. Extended Rotary Handle
8. Mechanical Interlock
9. Door Interlock
10. Terminal Covers
11. Padlocking Arrangement
12. Communication Module, where available

### 13. AUXILIARY CONTACT

Auxiliary contacts provide MCCB status feedback to the control circuit.

Typical status signals include:

- MCCB ON
- MCCB OFF

Auxiliary contacts can be connected to:

- PLC
- Relay
- SCADA
- Annunciator
- Control Panel
- ATS Controller

The exact contact configuration depends on the MCCB manufacturer.

### 14. ALARM CONTACT

An alarm contact is used to indicate a trip condition.

It can be used to distinguish between:

- Normal OFF condition
- Protection TRIP condition

The alarm signal can be connected to a PLC, relay, annunciator or SCADA system.

## 15. SHUNT TRIP

A Shunt Trip is an electrical accessory used to remotely trip or open the MCCB.

A trip command can be generated from:

- Emergency Stop
- Fire Alarm System
- Protection Relay
- PLC
- ATS Controller
- Remote Trip Push Button

The Shunt Trip operating voltage must match the selected accessory specification.

## 16. UNDER VOLTAGE RELEASE

Under Voltage Release is commonly known as UVR.

It is used to trip the MCCB or prevent closing when the control voltage falls below the specified operating level.

The exact operation depends on the selected MCCB and UVR accessory.

## 17. MOTORIZED MCCB

A motorized MCCB is provided with a motor operator for electrical opening and closing.

Motorized MCCBs are commonly used in:

- Automatic Changeover Systems
- ATS Systems
- Generator Changeover
- Remote Switching
- PLC-Controlled Distribution Systems
- Building Management Systems
- Remote Control Applications

The motor operator allows the MCCB to be controlled remotely or automatically.

## 18. WHAT IS A CHANGEOVER SYSTEM?

A changeover system transfers an electrical load from one power source to another power source.

Typical sources include:

- Mains-1
- Mains-2
- Utility Supply
- Generator
- Transformer-1
- Transformer-2

The changeover system ensures that the load is connected to the selected available source.

## 19. TYPES OF MCCB CHANGEOVER

Common MCCB changeover arrangements include:

1. Manual Changeover
2. Electrical Changeover
3. Automatic Changeover
4. Mains-to-Mains Changeover
5. Mains-to-DG Changeover
6. Transformer-to-Transformer Changeover
7. Two-Incomer Changeover
8. Incomer and Bus Coupler Arrangement
9. Motorized MCCB Changeover
10. ATS-Based Changeover

## 20. MANUAL MCCB CHANGEOVER

In a manual changeover arrangement, the operator manually changes the electrical supply from one source to another.

Typical arrangement:

SOURCE-1 → MCCB-1 → LOAD

SOURCE-2 → MCCB-2 → LOAD

For a normal non-parallel system, only one source should be connected to the load at a time.

A mechanical interlock can be provided between the two MCCBs to prevent simultaneous closing.

## 21. ELECTRICAL MCCB CHANGEOVER

In an electrical changeover arrangement, MCCBs are operated using electrical accessories.

Typical components include:

- Motor Operator
- Auxiliary Contacts
- Shunt Trip
- Under Voltage Release
- Control Relay
- Timer
- Selector Switch

- Control Circuit

The control circuit determines which MCCB can be closed.

## 22. AUTOMATIC MCCB CHANGEOVER

Automatic changeover transfers the load automatically from the failed source to the healthy source.

Typical arrangement:

SOURCE-1 → MCCB-1 → LOAD

SOURCE-2 → MCCB-2 → LOAD

An automatic transfer controller monitors the source conditions.

If Source-1 fails:

1. Source-1 failure is detected.
2. MCCB-1 is opened or confirmed OFF.
3. Required transfer delay is completed.
4. Source-2 health is verified.
5. MCCB-2 closing permissive is checked.
6. MCCB-2 is closed.
7. Load is transferred to Source-2.

### 23. MAINS-TO-MAINS CHANGEOVER

Mains-to-Mains changeover uses two independent utility sources.

SOURCE-1 → MCCB-1 → LOAD

SOURCE-2 → MCCB-2 → LOAD

Under normal operation, the selected source supplies the load.

If the selected source fails, the control system transfers the load to the alternate healthy source.

### 24. MAINS-TO-DG CHANGEOVER

Mains-to-DG changeover is commonly used where a diesel generator provides standby power.

Typical arrangement:

UTILITY / MAINS → MCCB-1 → LOAD

DG → MCCB-2 → LOAD

NORMAL CONDITION



Mains is healthy.

MCCB-1 = ON

MCCB-2 = OFF

Load is supplied from Mains.

## MAINS FAILURE

1. Mains failure is detected.
2. DG start command is generated.
3. DG starts.
4. DG voltage and frequency become healthy.
5. Mains MCCB is opened or confirmed OFF.
6. DG MCCB closing permissive is obtained.
7. DG MCCB closes.
8. Load is supplied by DG.

## 25. MAINS RESTORATION

When the mains supply returns:

1. Mains voltage is monitored.
2. Mains remains healthy for the required return delay.
3. DG MCCB opens.

4. Mains MCCB closing permissive is checked.
5. Mains MCCB closes.
6. Load is transferred back to Mains.
7. DG cool-down sequence may operate.
8. DG stops after the required sequence.

Exact timing depends on the project control philosophy.

## 26. TWO-INCOMER MCCB ARRANGEMENT

A two-incomer system can consist of two MCCBs.

INCOMER-1:

SOURCE-1 → MCCB-1 → BUS-1

INCOMER-2:

SOURCE-2 → MCCB-2 → BUS-2

A Bus Coupler MCCB can be provided between Bus-1 and Bus-2.

## 27. MCCB BUS COUPLER ARRANGEMENT

Typical arrangement:

SOURCE-1 → MCCB-1 → BUS-1

BUS-1 → BUS COUPLER MCCB → BUS-2

SOURCE-2 → MCCB-2 → BUS-2

The Bus Coupler can be used to transfer supply from one bus section to another when the required conditions are satisfied.

## 28. NORMAL CONDITION – TWO INCOMER SYSTEM

Typical normal condition:

Incomer-1 MCCB = ON

Incomer-2 MCCB = ON

Bus Coupler MCCB = OFF

Each source supplies its respective bus section.

## 29. SOURCE-1 FAILURE

If Source-1 fails:

1. Source-1 unhealthy condition is detected.

2. Incomer-1 MCCB is opened or confirmed OFF.
3. Source-2 must be healthy.
4. Incomer-2 must be ON and available.
5. Bus-2 must be healthy.
6. Bus Coupler closing permissives are checked.
7. Bus Coupler MCCB closes.
8. Bus-1 is supplied from Source-2.

### 30. SOURCE-2 FAILURE

If Source-2 fails:

1. Source-2 unhealthy condition is detected.
2. Incomer-2 MCCB is opened or confirmed OFF.
3. Source-1 must be healthy.
4. Incomer-1 must be ON and available.
5. Bus-1 must be healthy.
6. Bus Coupler closing permissives are checked.
7. Bus Coupler MCCB closes.
8. Bus-2 is supplied from Source-1.

### 31. ELECTRICAL INTERLOCKING

Electrical interlocking prevents incorrect MCCB operation.

Typical interlocks include:

- MCCB-1 ON prevents MCCB-2 closing.
- MCCB-2 ON prevents MCCB-1 closing.
- Unhealthy source prevents closing.
- Faulted MCCB prevents closing.
- Emergency trip prevents closing.
- Earth switch ON prevents closing where applicable.
- Maintenance condition prevents automatic operation.

The final interlocking philosophy shall be based on the approved project design.

## 32. MECHANICAL INTERLOCKING

Mechanical interlocking physically prevents two MCCBs from being closed simultaneously.

Example:

MCCB-1 CLOSED



MCCB-2 CLOSE NOT PERMITTED

MCCB-2 CLOSED



## MCCB-1 CLOSE NOT PERMITTED

Mechanical interlocking is especially important in non-parallel changeover systems.

### 33. MCCB CHANGEOVER WITH MOTOR OPERATOR

For automatic changeover applications, MCCBs may be equipped with motor operators.

Typical components include:

- MCCB-1
- MCCB-2
- Motor Operators
- Auxiliary Contacts
- Alarm Contacts
- Shunt Trips
- UV Releases, where required
- ATS Controller
- Control Relays
- Timers
- Electrical Interlock
- Mechanical Interlock

### 34. ATS – AUTOMATIC TRANSFER SWITCH

ATS stands for Automatic Transfer Switch.

An ATS automatically transfers the electrical load between two available power sources.

A typical ATS system may consist of:

- Source-1 Monitoring
- Source-2 Monitoring
- Transfer Controller
- MCCB or ACB
- Motor Operators
- Auxiliary Contacts
- Electrical Interlocking
- Mechanical Interlocking
- Timers
- Control Relays

### 35. ATS SOURCE MONITORING

The ATS controller may monitor:

- Phase Voltage
- Under Voltage
- Over Voltage
- Phase Failure
- Phase Sequence
- Frequency

- Source Availability

The exact monitoring functions depend on the selected ATS controller.

### 36. SOURCE HEALTHY CONDITION

A source is considered healthy when the required electrical parameters are within the configured limits.

Typical parameters include:

- Voltage available
- Correct phase sequence
- Frequency within acceptable range
- No phase failure
- No configured source fault

Exact healthy limits must be taken from the approved controller settings and project specification.

### 37. CHANGEOVER CONTROLLER

The changeover controller receives source and breaker status signals and generates breaker control commands.

Typical inputs include:



- Source-1 Healthy
- Source-2 Healthy
- MCCB-1 ON
- MCCB-1 OFF
- MCCB-1 TRIP
- MCCB-2 ON
- MCCB-2 OFF
- MCCB-2 TRIP
- Emergency Stop
- Auto/Manual Selection

Typical outputs include:

- MCCB-1 Close
- MCCB-1 Trip/Open
- MCCB-2 Close
- MCCB-2 Trip/Open
- DG Start
- Alarm

### 38. AUTO / MANUAL SELECTOR SWITCH

A selector switch can be used to select the operating mode.

AUTO:

Automatic changeover operation is enabled.

MANUAL:

Manual operation is selected and automatic transfer operation is disabled according to the control philosophy.

### 39. AUTOMATIC CHANGEOVER SEQUENCE

Typical sequence:

NORMAL CONDITION

Source-1 = Healthy

MCCB-1 = ON

MCCB-2 = OFF

SOURCE-1 FAILURE

Source-1 becomes unhealthy.

TRANSFER SEQUENCE

MCCB-1 opens or is confirmed OFF.

## SOURCE-2 CHECK

Source-2 health is verified.

## CLOSE

MCCB-2 closes.

## LOAD TRANSFERRED

Load is supplied from Source-2.

## 40. RETURN TRANSFER

When Source-1 becomes healthy again:

1. Source-1 health is confirmed.
2. Return delay timer operates.
3. MCCB-2 opens.
4. MCCB-1 closing permissive is checked.
5. MCCB-1 closes.
6. Load returns to Source-1.

This operation is commonly called Return Transfer or Retransfer.

#### 41. OPEN TRANSITION

In an open-transition changeover:

Source-1 is disconnected before Source-2 is connected.

Sequence:

MCCB-1 OFF



Transfer Delay



MCCB-2 ON

This arrangement prevents simultaneous connection of both sources.

#### 42. CLOSED TRANSITION

In a closed-transition system, both sources may be temporarily connected during transfer under controlled synchronization and approved paralleling conditions.

This requires suitable:

- Synchronization
- Protection
- Interlocking
- Paralleling Controls

A normal non-parallel changeover system must not be used for closed-transition operation without suitable engineering.

#### 43. BREAK-BEFORE-MAKE

Break-before-make means that the first source is disconnected before the second source is connected.

Example:

MCCB-1 OFF



Transfer Delay



MCCB-2 ON

This arrangement is commonly used where the two sources must not be paralleled.

#### 44. MAKE-BEFORE-BREAK

Make-before-break means the second source is connected before the first source is disconnected.

This requires a specially engineered system with suitable synchronization, protection and paralleling controls.

It should only be used when the sources are specifically designed and approved for parallel operation.

#### 45. MCCB CHANGEOVER PERMISSIVES

Before closing an MCCB during automatic changeover, the following conditions may be checked:

1. Correct operating mode selected.
2. Required source healthy.
3. Other source breaker OFF.
4. No fault-trip condition.
5. MCCB available for closing.
6. Motor operator/control supply healthy.
7. Required auxiliary feedback available.
8. Mechanical interlock healthy.
9. Electrical interlock healthy.

10. Emergency trip not active.

#### 46. MCCB STATUS SIGNALS

Typical MCCB signals are:

MCCB ON

MCCB OFF

MCCB TRIP

FAULT TRIP

READY

LOCAL

REMOTE

CLOSE COMMAND

TRIP COMMAND

SPRING CHARGED, where applicable

MOTOR OPERATOR HEALTHY, where applicable

#### 47. MCCB CHANGEOVER CONTROL COMPONENTS

A typical automatic MCCB changeover panel may contain:

1. MCCB-1

2. MCCB-2

3. Motor Operators

4. Auxiliary Contacts

5. Alarm Contacts
6. Shunt Trips
7. UV Releases, where required
8. ATS Controller
9. Control Relays
10. Timers
11. Control MCB/Fuse
12. Auto/Manual Selector Switch
13. Indicating Lamps
14. Push Buttons
15. Terminal Blocks
16. Mechanical Interlock
17. Electrical Interlock

#### 48. INDICATION LAMPS

Typical indications include:

SOURCE-1 AVAILABLE

Indicates Source-1 is healthy.

SOURCE-2 AVAILABLE

Indicates Source-2 is healthy.

MCCB-1 ON



Indicates MCCB-1 is closed.

MCCB-1 TRIP

Indicates MCCB-1 has tripped.

MCCB-2 ON

Indicates MCCB-2 is closed.

MCCB-2 TRIP

Indicates MCCB-2 has tripped.

AUTO MODE

Indicates automatic mode is selected.

MANUAL MODE

Indicates manual mode is selected.

COMMON FAULT

Indicates a common fault/alarm condition.

#### 49. MCCB SELECTION PROCEDURE

The MCCB should be selected based on:

1. Load Current
2. System Voltage
3. Frequency
4. Number of Poles
5. Short-Circuit Current
6. Breaking Capacity
7. Cable Ampacity
8. Installation Conditions
9. Ambient Temperature
10. Protection Coordination
11. Selectivity Requirements
12. Required Accessories
13. Changeover Requirements

#### 50. MCCB CURRENT SELECTION

The MCCB rated current should be suitable for the connected load and cable system.

A basic design relationship is:

$$\text{Load Current} \leq \text{MCCB Rating} \leq \text{Cable Permissible Current}$$

However, actual selection must also consider:

- Starting Current
- Ambient Temperature
- Cable Grouping
- Installation Method
- Protection Settings
- Coordination
- Manufacturer Recommendations

## 51. MCCB FOR MOTOR LOAD

For motor applications, MCCB selection should consider:

- Motor Full Load Current
- Starting Current
- Starting Method
- Motor Protection
- Cable Size
- Short-Circuit Level
- Coordination with Contactor
- Overload Relay Settings

The MCCB should not be selected only from the motor nameplate current.

## 52. MCCB FOR TRANSFORMER FEEDER

Transformer feeder MCCB selection should consider:

- Transformer Rating
- Primary Voltage
- Secondary Voltage
- Full Load Current
- Short-Circuit Level
- Transformer Inrush Current
- Cable/Busbar Rating
- Protection Coordination

### 53. MCCB FOR GENERATOR FEEDER

Generator feeder MCCB selection should consider:

- Generator Rated Current
- Generator Short-Circuit Capability
- Voltage
- Frequency
- Neutral Arrangement
- Protection
- ATS Requir